

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Sixth Semester of B. Tech (EE) Examination****May 2018****EE306/EE306.01 Electrical Machines III****Date: 01.05.2018, Tuesday****Time: 10.00 a.m. To 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer following questions. [07]**

- (1) What parameters can be known from the circle diagram of synchronous motor?
- (2) What do you mean by load angle of synchronous motor?
- (3) Enumerate the permanent magnet materials used for permanent magnet motor.
- (4) State the applications of linear synchronous motor.
- (5) State the advantages and disadvantages of synchronous condenser.
- (6) Draw the connection diagram of three stage cryogenerator.
- (7) State the similarities between conventional synchronous motor and PMSM.

Q – 2.a Explain the experimental setup to draw the V curve of synchronous motor. [07]**Q – 2.b A 3- Φ , 400V synchronous motor operates at full load to give 50kW output with 92% efficiency, if field current is adjusted to make its power factor 0.8 leading, estimate the armature current at same motor efficiency. [04]****Q – 2.c Explain in brief, the starting of synchronous motor under load condition. [03]****OR****Q – 2.a Write a short note on damper winding and surging of a synchronous motor. [07]****Q – 2.b A 3- Φ , 50Hz, star connected synchronous motor rated for 500kVA, 3.3kV, 87A, 300rpm, has an effective armature resistance and synchronous reactance per phase of 1Ω and 10Ω respectively. Determine the internal mechanical power developed by motor when it is operating at rated current and 0.866 power factor leading. Also calculate no of poles. [07]****Q – 3 Answer Any Two. (Each question carry seven marks) [14]**

- (1) Draw and explain speed and torque control mechanism through feedback loop in Permanent Magnet Brushless DC motor.
- (2) Explain the design consideration and construction of PMDC motor.

- (3) A 10hp output, 415V, 3- Φ , star connected synchronous motor has negligible armature resistance and synchronous reactance of 5 Ω /phase. Determine minimum armature current and corresponding excitation voltage for full load condition. Assume $\eta = 90\%$.

SECTION – II

- Q - 4** Define the armature reaction. Explain its effect on alternator delivering lagging power factor load with necessary diagrams. [07]
- Q – 5.a** Explain the procedure to perform the slip test on salient pole alternator. Show the calculation of X_d and X_q with appropriate waveforms. [07]
- Q – 5.b** Define and explain the pitch factor and distribution factor. Derive the emf equation of alternator. [07]

OR

- Q – 5.a** Differentiate between turbo generators and hydro generators. [07]
- Q – 5.b** Derive the active, reactive and apparent output power equations of salient pole synchronous generator. [07]
- Q – 6. Answer Any Two. (Each question carry seven marks)** [14]
- (1) Explain the capability curve of alternator.
- (2) State the conditions for synchronizing the two alternators. Explain the one dark and two bright lamp method.
- (3) Define the voltage regulation of alternator. Write the steps for find the voltage regulation by zero power factor (ZPF) method. Show how to calculate X_l and X_a from Potier triangle.
